MSTNet: A Multilevel Spectral–Spatial Transformer Network for Hyperspectral Image Classification

Haoyang Yu, *Member, IEEE*, Zhen Xu, Ke Zheng, Danfeng Hong, *Senior Member, IEEE*, Hao Yang, and Meiping Song

***Abstract*— Convolutional neural networks (CNNs) have been widely used in hyperspectral image classification (HSIC). Although the current CNN-based methods have achieved good performance, they still face a series of challenges. For example, the receptive field is limited, information is lost in down-sampling layer, and a lot of computing resources are consumed for deep networks. To overcome these problems, we proposed a multilevel spectral–spatial transformer network (MSTNet) for HSIC. The structure of MSTNet is an image-based classification framework, which is efficient and straightforward. Based on this framework, we designed a self-attentive encoder. First, HSIs are processed into sequences. Meanwhile, a learned positional embedding (PE) is added to integrate spatial information. Then, a pure transformer encoder (TE) is employed to learn feature representations. Finally, the multilevel features are processed by decoders to generate the classification results in the original image size. The experimental results based on three real hyperspectral datasets demonstrate the efficiency of the proposed method in comparison with the other related CNN-based methods.**

***Index Terms*— Convolutional neural networks (CNNs), hyperspectral image (HSI), image-based classification, transformer.**

I. Introduction

HYPERSPECTRAL image (HSI) is a 3-D data, which is composed of hundreds of continuous spectral bands. It contains abundant spectral information, which can extract detailed features of different ground objects [1], [2]. HSI classification (HSIC) aims to assign a unique label to each pixel in the scene, so that it can be identified and defined by a given class. Recently, HSIC has become research hot spot, which has been widely used in precision agriculture, environment monitoring, and mineral exploration [3]–[7]. In the early research, many machine learning methods are introduced for the HSIC, such as the support vector machine (SVM) [8]–[10], decision tree classification, and maximum likelihood estimate. However, these traditional methods are restricted since they only extract low-level features and ignore high-level features, thus failing to achieve higher accuracies. Also, HSI contains a lot of redundant information, which poses a series of challenge for the classifiers [11]–[13].

In recent years, deep learning-based models have also been introduced for the HSIC due to its excellent ability of features extraction [14]–[17]. Chen *et al.* [18] introduced deep learning to HSIC for the first time and extracted the spectral features of HSI through an automatic encoder. Compared with the machine learning methods, the classification accuracy is significantly improved, but this method only considers the spectral information and ignores the spatial information. After that, convolution neural network (CNN) is widely used in HSIC. Spectral–spatial features can be extracted using a deep CNN (DCNN), and these features represent low- to high-level semantic information. Makantasis *et al.* [19] proposed a feature learning approach for HSIC based on CNNs, which can learn structured features directly from the hyperspectral input data. Yue *et al.* [20] proposed a novel deep learning framework for HSIC using both spectral and spatial features, which includes principal component analysis (PCA), DCNNs, and logistic regression (LR). Li *et al.* [21] proposed a novel pixel-pair method to significantly increase the number of training samples, which gives full play to the advantage of CNN. Zhong *et al.* [22] proposed an end-to-end spectral–spatial residual network (SSRN) that takes raw 3-D cubes as input data without feature engineering for HSIC.

More recently, various CNNs with multilevel and multiscale spectral–spatial features have been introduced for HSIC. Haque and Mishu [23] proposed a hybrid approach of a dimensionality reduction technique based on PCA and a novel multiscale CNN named PCA-MS-CNN in order to extract spectral and spatial features for HSI. He *et al.* [24] proposed a multiscale 3-D DCNN (M3D-DCNN) for HSIC, which could jointly learn both 2-D multiscale spatial features and 1-D spectral features from HSI data in an end-to-end approach. Hang *et al.* [25] proposed an attention aided CNN model for spectral–spatial classification of HSIs. Specifically, a spectral attention subnetwork and a spatial attention subnetwork are proposed for spectral and spatial classification, respectively. Yu *et al.* [26] proposed a dual-channel convolution network (DCCN) for the global and multiscale information utilization of HSI data. Cao *et al.* [27] proposed an interpretable deep

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network structure for pan sharpening, which has clear physical interpretations, and all the learnable modules can be automatically learned in an end-to-end way from the given dataset. Lee and Kwon [28] proposed a contextual deep CNN, which can optimally explore local contextual interactions by jointly exploiting local spectral–spatial relationships of neighboring individual pixel vectors. Hong *et al.* [29] proposed a two-stream Siamese deep network, which learns an additional network from the pure or nearly pure endmembers to correct the weights of another unmixing network by sharing network parameters and adding spectrally meaningful constraints (e.g., nonnegativity and sum-to-one) toward a more accurate and interpretable unmixing solution. Bei *et al.* [30] proposed a novel end-to-end 3-D dense CNN with spectralwise attention mechanism (SA) for HSIC. Hang *et al.* [31] proposed a multitask generative adversarial network (MTGAN) to alleviate this issue by taking advantage of the rich information from unlabeled samples. Specifically, we design a generator network to simultaneously undertake two tasks: the reconstruction task and the classification task. Cao *et al.* [32] proposed deep spatial–spectral global reasoning network to consider both the local and global information for HSI noise removal. Hong *et al.* [33] proposed a baseline solution to the aforementioned difficulty by developing a general multimodal deep learning (MDL) framework. Gao *et al.* [34] proposed a densely connected multiscale attention network (DMSAN) based on 3-D convolution for HSIC, which solves the problem of difficult extraction of features and underutilization of information from HSI data.

Although CNN-based models have been proven its effectiveness for HSIC, they still exist some problems. First, it is difficult for CNN-based methods to take into account global information in HSI. In general, the efficiency of feature extraction is dependent on the convolutional kernels of each layer, but the number of parameters increases due to the deepening of the network, and convolutional kernels are set to 3 or 5 to ensure the balance between the number of network layers and computational efficiency. Thus, the CNN receptive field is restricted by the small kernel size. Second, it is difficult for patch-based classification framework for HSIC to achieve a fast calculation process. A lot of redundant information will be generated by using patch as input leads to more computational resources are occupied. Moreover, patch-based classification framework limits the understanding of the remote sensing images. Finally, it is difficult for CNN to build a lightweight and efficient classification network. In most CNN-based models for HSIC, deep networks could often achieve better classification results, but it also causes the network to be more complex.

To overcome the above problems of CNN-based methods, we focused on the transformer, which is a natural language processing (NLP) model. Transformer draws global dependency within a sequence of input by self-attention (SA). Recently, some transformer-based models are used in image processing. Dosovitskiy *et al.* [35] proposed a pure transformer applied directly to the sequences of image patches for classification, called vision transformer (ViT). ViT proves that it is unnecessary to rely on CNN in the visual task. Carion *et al.* [36] proposed a new method that views object detection as a direct set prediction problem, called DEtection TRansformer or DETR, which reasons about the relations of the objects and the global image context to directly output the final set of predictions in parallel. It can be seen that we can convert the image into a sequence to adapt to the input of the transformer for feature extraction. Hong *et al.* [37] proposed a new class of backbone network SpectralFormer based on the sequential perspective of the transformer, which can adapt to pixel and small-block inputs. Compared with CNN-based methods, transformer has the following advantages. First, the receptive field is not restricted by convolutional kernels because the entire image can be converted into a vector as an input. Second, feature maps always maintain the same resolution during the feature extraction process to avoid losing effective information. Finally, training resources are reduced effectively to improve computing efficiency.

In this article, we propose a novel perspective, which a transformer-based network is used to replace the traditional CNN to perform HSIC tasks. The proposed network incorporates an image-based classification framework for HSI to extract richer global information. The proposed method, named multilevel spectral–spatial transformer network (MSTNet), includes the contributions as follows.

1) The traditional CNN is replaced with the transformer-based model, which overcomes the problem of limited receptive fields and makes the feature extraction process have a global view.
2) The traditional patch-based framework is replaced with the image-based classification framework, which solve the inefficiency in the training and testing. The proposed framework is more efficient and straightforward.
3) The proposed MSTNet is lightweight to improve computational efficiency, which can also flexibly adapt to arbitrary size image as an input. Moreover, multilevel feature results are fused to improve the representation ability for HSI.

In the remainder of this article, Section II presents the proposed method. Section III evaluates the classification performance. Section IV concludes this article.

II. MSTNet

In this section, we will introduce the individual components included in the proposed method. On this basis, we proposed the MSTNet to finish classification tasks for HSI.

A. Image-Based Classification Framework for HSI

The traditional deep learning HSIC framework is a patch-based framework. The training patch is composed of training pixels and its neighboring pixels, and then, the patch is input into the model and the label of the center pixel of the patch is predicted. This method generates patches pixel by pixel during testing. Then, the patches are imported into the model one by one. Finally, the predicted labels are reshaped into an image of the same size as the original image.

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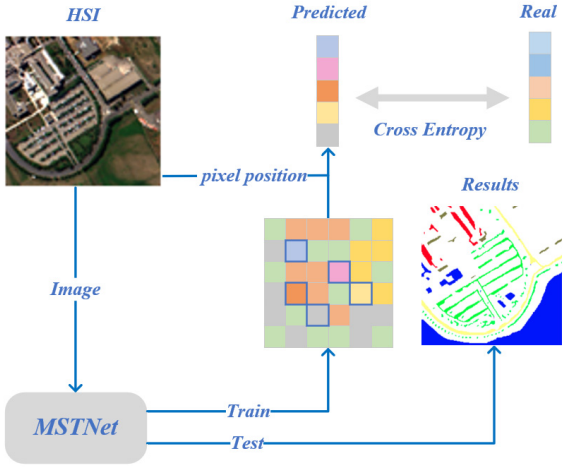


Fig. 1. Image-based classification framework for HSI.

Obviously, the patch-based classification method has some disadvantages. First, the receptive field of the model is limited by the patch size. Second, the model needs to be redesigned if the patch size is changed. The optimal patch size is related to the image spatial resolution, making it difficult to design a general classification model. Finally, during the testing phase, the pixel-by-pixel patch generation method results in information redundancy, consumes lots of computing resources, and has low time efficiency.

In this article, we introduced an image-based classification framework (see Fig. 1), which can overcome the above issues associated with patch-based classification. Arbitrary semantic segmentation model can be applied to this framework for HSIC. During the training phase, input is the image containing training samples, and the predicted labels for all corresponding pixels are used as the output. We use the position of labeled samples as a mask, and it covers the output to select the corresponding pixels. During the testing phase, we bring the image into the model and then predict the corresponding labels to all pixels. Compared to the traditional patch-based classification frameworks, image-based classification frameworks have a faster inference process in testing since the information is nonoutput during each inference process, which can save a lot of computing resources.

B. CNN Layer

CNN layer contains two components. In the first component, a 1×1 convolution layer, a dropout layer, an instance normalization layer, and a nonlinearity activation layer are combined to integrate spectral information by changing the number of channels and increase and reduce dimension by changing the number of convolution kernels. Also, the proposed MSTNet follows the same input–output structure as in NLP for transformation between 2-D sequences. So, there is a mismatch between the 3-D image and the 2-D sequence. More specifically, transformer receives a 2-D sequence of feature embeddings $\mathrm{Z} \in \mathrm{R}^{L \times C}$ as an input, L is the length of sequence, and C is the channel size. Thus, a feature map $x \in \mathrm{R}^{H \times W \times C}$ needs to be converted into Z.

For a real HSI, Pavia University, it contains 610×340 pixels. The final sequence will have a length of 207 400. Considering that avoid the limitation of GPU memory. Therefore, a 5×5 convolutional layer as the second component to reduce the spatial size of the original image. Moreover, since the channel information was integrated before, the number of channels is compressed to C' in order to save computing resources. Finally, a feature map $x' \in \mathrm{R}^{H' \times W' \times C'}$ is generated through CNN layer.

C. Linear Projection

As mentioned before, 3-D images need to be processed into 2-D vectors to adapt to the input of the transformer. For image sequentialization, the feature map x' is flattened into a sequence $e \in \mathrm{R}^{L' \times C'}$, and L' is equal to H' times W'.

D. Positional Embedding

After image sequentialization, the dimension of the sequence e is matched for the transformer. However, the sequence e is disorderly lead to the loss of image spatial information. Thus, a unique embedding p for every pixel is learned and is added to the sequence e to generate final sequence input

$$E = e + p \tag{1}$$

which contains spatial information.

E. Transformer Encoder (TE)

The structure of transformer encoder (TE) is shown in Fig. 3. It includes two layer normalizations (LNs), a multihead SA (MSA), and a multilayer perceptron (MLP). In the training phase, the TE is executed four times to generate different level features.

1) Layer Normalization (LN): In general, normalization is performed before the training phase in order to standardize data distribution. If the distribution of training data and test data is different, the generalization ability of the network will be severely reduced. At present, the two normalization methods commonly used in deep learning are batch normalization (BN) and LN. In recent year, BN has shown good performance in the field of computer vision. However, BN leads to significant performance degradation for NLP tasks. Shen *et al.* [38] perform a systematic study of NLP transformer models and find that the statistics of NLP data across the batch dimension exhibit large fluctuations throughout training. Therefore, LN is introduced into TE to alleviate gradient disappearance and enhance model characterization capabilities. Also, LN has become the first choice to be used in NLP according to empirical observations.

2) Multihead Self-Attention (MSA): The SA is the key component of the transformer model. The structure of SA is shown in Fig. 2. The matrix Q(query), K(key value), and V(value) are used in the calculation process. In fact, SA receives embeddings E as an input, get Q(query), K(key value), and V(value) by linearly changing the embeddings, and

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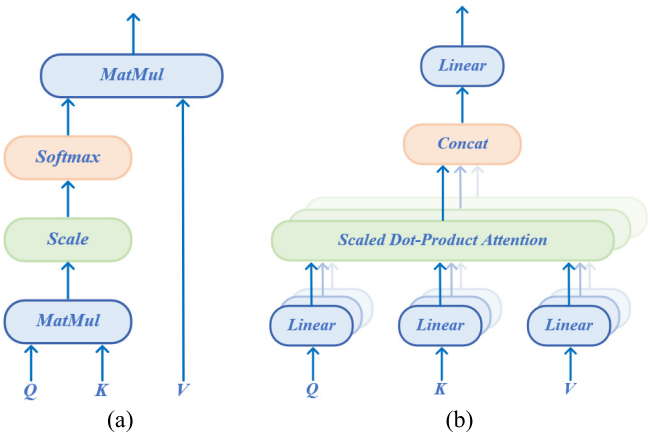


Fig. 2. Illustration of the attention mechanism in transformers. (a) SA. (b) Multihead attention.

then, the softmax function is used to compute the weights on the value. The specific calculation is as follows:

$$\text{Attention}(Q, K, V) = \text{soft max}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{2}$$

where d_k represents the dimension of Q and K.

Furtherly, the outputs of multiple independent SA components are concatenated. This process is called MSA. The architecture is shown in Fig. 2. Our MSA uses head and is written as

$$\text{MultiHead}(Q, K, V) = \text{concat}(\text{head}_1, \ldots, \text{head}_h)W^O \tag{3}$$

where W^O represents the learned parameter. Also, each *head* is expressed as

$$\text{head}_i = \text{Attention}\left(QW_i^Q, KW_i^K, VW_i^V\right) \tag{4}$$

where $W_i^Q \in \mathrm{R}^{C' \times d_q}$, $W_i^K \in \mathrm{R}^{C' \times d_k}$, and $W_i^V \in \mathrm{R}^{C' \times d_v}$.

3) Multilayer Perceptron (MLP): MLP consists of two fully connected layer and a nonlinearity named the Gaussian error linear unit (GELU) activation. GELU can effectively integrate inactive information, which is defined as

$$\text{GELU} = x\Phi(x) = x \cdot \frac{1}{2}\left[1 + \text{erf}\left(x/\sqrt{2}\right)\right] \tag{5}$$

where $\Phi(x)$ represents the standard Gaussian cumulative distribution function, $\text{erf}(x) = \int_0^x e^{-t^2} dt$.

F. Multilevel Feature Aggregation Decoder (MFAD)

After obtaining the learned feature results of the TE, we need to reshape the encoder's features from a 2-D embedding shape to a 3-D feature map of original spatial size in order to complete pixel-level segmentation. In decoder structures, multilevel feature aggregation is introduced to improve representation ability.

The multilevel feature aggregation decoder (MFAD) is shown in Fig. 3. Different levels of features are generated after calculation by the TE. The low-level feature maps pay more attention to the correlation of global pixels, while the high-level feature maps pay more attention to the correlation of local pixels. Through elementwise addition, the interaction among different feature maps can be improved, thereby improving the expressive ability of the network. The representation of multilayer TE features is taken as an input, and each layer of features is processed separately. Specifically, the encoder's feature is first reshaped from a 2-D embedding shape to a 3-D feature map. A 1×1 convolution layer and a 3×3 convolution layer are then applied to halve the feature channels, respectively. The spatial resolution of the feature map is next increased by four times through two upsampling layers. Also, a top-down aggregation constructor through elementwise addition is introduced after the first layer to improve the interactions across different level features. Multilevel features are finally connected together through channels to generate fusion features that are upsampled to the original spatial resolution.

G. Multilevel Spectral–Spatial Transformer Network (MSTNet)

Based on the above components, we propose an MSTNet. The network structure is shown in Fig. 3. Our design concept follows efficient, fast, and lightweight. The overall workflow and network structure are as follows.

First, an HSI is input into the MSTNet through the image-based classification framework. A feature map is then generated by CNN layer in order to integrate channel information and reduce spatial resolution. Second, a 3-D feature map is reshaped into a 2-D sequence containing spatial information using linear projection and positional embedding (PE) to match the input size of the transformer. TE next takes the feature sequence as an input to learn the association between different pixels through the MSA mechanism and output the learned feature results in the MLP layer. Especially, we set the network to four layers to generate different level feature results. Finally, MFAD receives feature maps from TE and produces multilevel fusion feature maps through a channel addition strategy.

The final output feature is sliced into a vector according to the position of training samples following the proposed image-based classification framework described Section II-A. The loss between the vector labels and the corresponding true labels is calculated using cross entropy [39].

In MSTNet, the dimension of the feature is set to 16 after image sequentialization in order to avoid excessive redundant channel information and effectively improve the computational efficiency. In addition, the number of TE layer is set to 4, so that it meets an effective classification result in the efficient calculation process. The number of head in MSA is set to 2, which is the best result of our many experiments.

III. Experiments and Analysis

In this section, the proposed MSTNet is evaluated using three real hyperspectral datasets, of which the details are introduced in Section III-A. The related parameter settings are reported in Section III-B. For comparative purposes, several well-established models, such as radial basis function (RBF)-SVM [40], 3DCNN [41], SSRN [22], adaptive spectral–spatial

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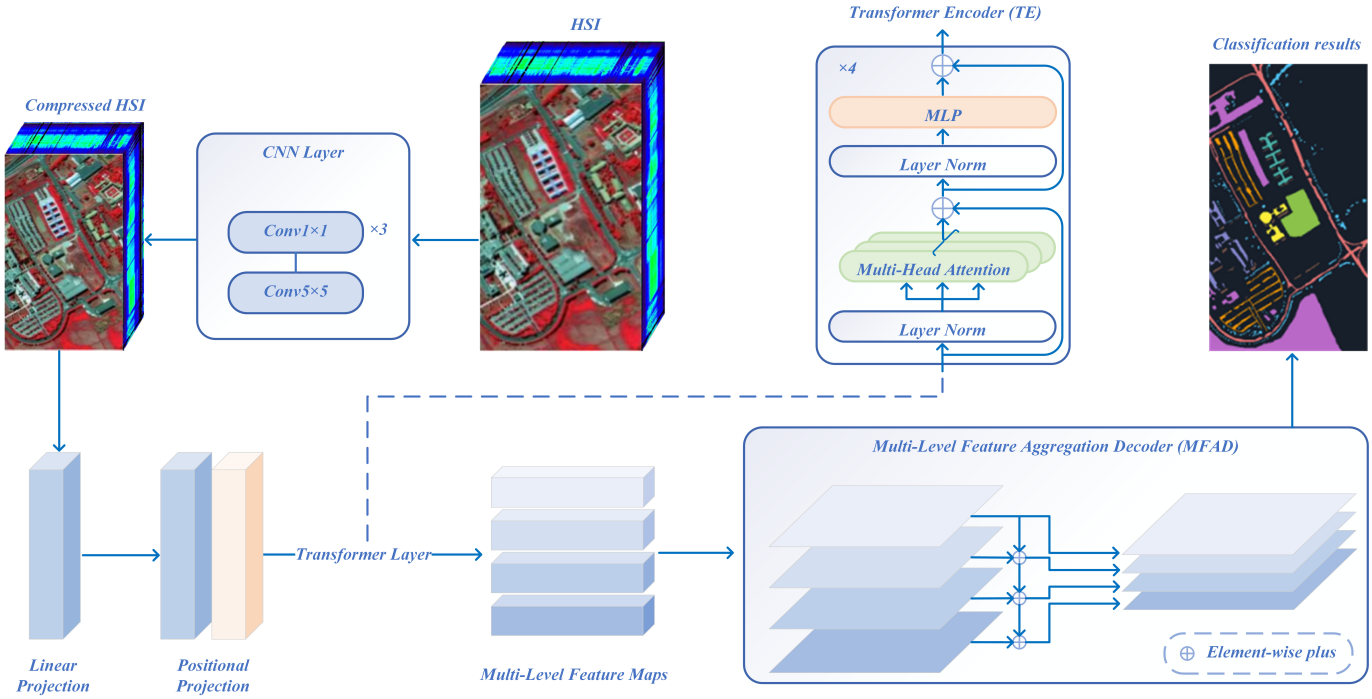


Fig. 3. Overall architecture of the proposed MSTNet for HSIC. MSTNet consists of a CNN layer, two projection layer, transformer layer (TE), and MFAD.

kernel ResNet (A^2S^2K-ResNet) [42], double-branch dual-attention (DBDA) [43], LinkNet [44], and UNet [45], are selected. The corresponding analysis and discussion of the classification results are reported in Section IV-C.

A. Data Description

Three HSIs were used to evaluate the performance of the proposed MSTNet including Pavia University, Indian Pines, and Salinas.

1) Indian Pines Dataset: The first data are the Indian Pines scene captured by the AVIRIS sensor over the Indian Pines region in Northwestern Indiana. The spatial size of it is 145 × 145 with the spatial resolution of 20 m. The number of spectral bands is 220 with the spectral coverage ranging from 0.4 to 2.5 μm. However, 20 bands are removed due to the high noise. The reference of this image includes 16 ground-truth classes, with a total of 10 249 labeled samples. The false-color composite image and the available ground-truth map are shown in Fig. 4.

2) University of Pavia Dataset: The second data are the University of Pavia scene collected by the ROSIS sensor over an urban area surrounding the University of Pavia, Italy. The spatial size of it is 610 × 340 with the spatial resolution of 1.3 m. The number of spectral bands is 115 with the spectral coverage ranging from 0.43 to 0.86 μm. However, 12 bands are removed due to the high noise. The reference of this image includes nine mutually exclusive classes, with a total of 42 776 labeled samples. The false-color composite image and the available ground-truth map are shown in Fig. 5.

3) Salinas Dataset: The last data are the Salinas dataset required by the AVIRIS sensor over Salinas Valley, CA, USA. The spatial size of it is 512 × 217 with the spatial resolution of 3.7 m. The number of spectral bands is 224 with the spectral coverage ranging from 0.2 to 2.4 μm. However, 20 bands are removed due to the high noise. The reference of this image includes 16 ground-truth classes, with a total of 54 129 labeled samples. The true-color composite image and the available ground-truth map are shown in Fig. 6.

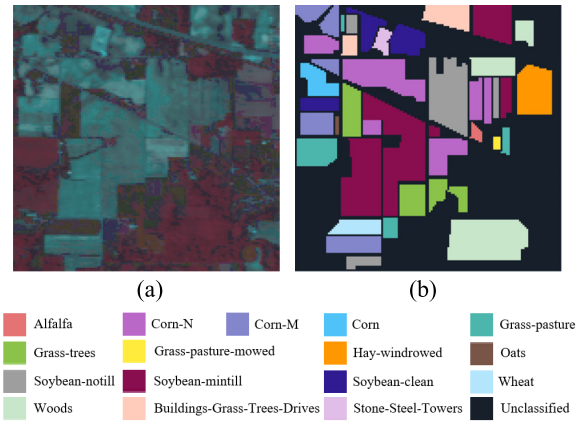


Fig. 4. AVIRIS indian pines scene. (a) True-color composite image. (b) Reference map.

B. Parameter Setting

The Kaiming weight initialization method [46] was adopted to initialize the parameters of the proposed network. Taking into account of the convergence speed and convergence accuracy, the network was trained using an Adam optimizer [47] with the default parameters set as: $\beta 1 = 0.9$, $\beta 2 = 0.999$, and $\beta 3 = 10^{-8}$. Considering the training stability, the initial

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TABLE I

OVERALL, AVERAGE, K STATISTIC, AND INDIVIDUAL CLASS ACCURACIES FOR THE INDIAN PINES DATA WITH 50 TRAINING SAMPLES PER CLASS. THE HIGHEST ACCURACIES ARE HIGHLIGHTED IN BOLD

Class	RBF-SVM	3DCNN	SSRN	DBDA	A^2S^2K-ResNet	LinkNet	UNet	MSTNet
1	78.90%	88.24%	97.87%	94.85%	**100.00%**	76.67%	80.70%	82.14%
2	61.14%	70.96%	90.32%	90.82%	85.56%	81.53%	86.77%	**99.20%**
3	57.27%	72.73%	86.19%	82.38%	86.97%	87.69%	87.99%	**98.20%**
4	57.84%	80.20%	93.05%	93.25%	90.25%	**99.14%**	82.29%	95.18%
5	84.03%	91.94%	91.55%	90.24%	96.69%	97.49%	94.62%	**98.34%**
6	92.26%	98.50%	99.12%	96.22%	99.59%	97.35%	99.04%	**99.17%**
7	79.41%	90.32%	**100.00%**	91.80%	**100.00%**	96.55%	**100.00%**	96.55%
8	97.99%	97.54%	97.42%	97.42%	99.90%	**100.00%**	98.76%	99.79%
9	62.96%	83.33%	**100.00%**	93.02%	95.24%	40.82%	86.95%	**100.00%**
10	66.14%	71.77%	82.77%	85.36%	85.93%	**92.50%**	89.81%	89.98%
11	63.20%	71.86%	87.19%	90.12%	85.46%	96.52%	95.91%	**99.14%**
12	68.79%	71.26%	84.59%	87.46%	85.94%	88.26%	**94.12%**	93.69%
13	94.01%	95.79%	**100.00%**	95.35%	**100.00%**	**100.00%**	97.61%	98.55%
14	89.65%	95.08%	98.38%	98.07%	98.06%	98.67%	**98.81%**	97.81%
15	58.28%	72.39%	77.89%	77.99%	83.92%	**98.21%**	94.37%	93.91%
16	93.33%	**96.84%**	92.08%	85.71%	95.88%	78.81%	86.11%	93.00%
OA	71.49%	79.24%	87.69%	88.07%	88.87%	92.71%	93.24%	**97.18%**
AA	75.33%	84.30%	92.40%	90.62%	93.09%	89.39%	92.12%	**95.91%**
Kappa	68.00%	76.67%	86.11%	86.54%	87.45%	91.73%	92.33%	**96.80%**

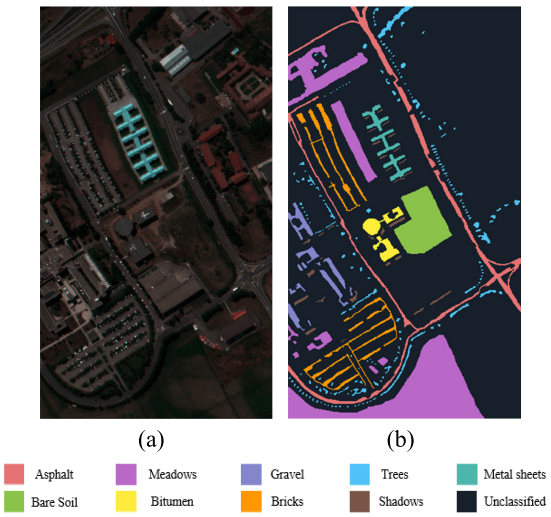


Fig. 5. ROSIS university of pavia scene. (a) True-color composite image. (b) Reference map.

learning rate was set as 5×10^{-4} after selecting the optimal parameter. The learning rate decay was 0.9 per 500 epochs and the total epochs was 2000. The batch size was set to one since only one input image was used in each experiment. The PyTorch deep learning framework [48] was used to train the proposed MSTNet. The computing environment consisted of the following specifications: i7-7820X CPU, 32 GB of RAM, and a GTX 2080Ti 11-GB GPU.

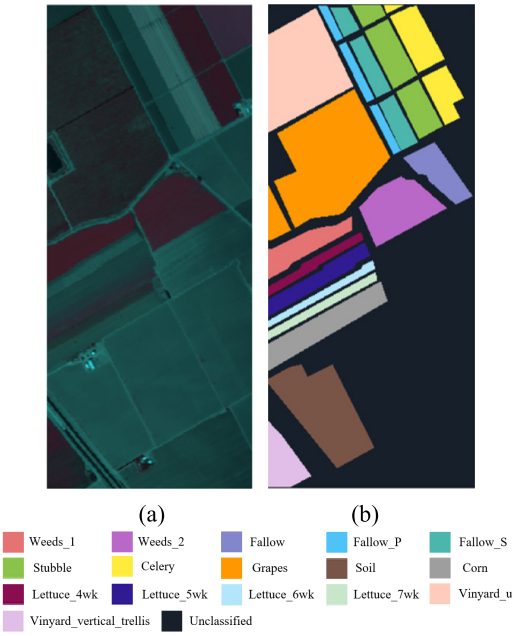


Fig. 6. AVIRIS salinas scene. (a) True-color composite image. (b) Reference map.

C. Experimental Results and Analysis

1) Indian Pines Scene: In the first test, the AVIRIS Indian Pines Scene was used to compare the performance of different classification methods. We selected 50 samples per class

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TABLE II

OVERALL CLASSIFICATION ACCURACIES (IN PERCENTAGE) OBTAINED BY THE DIFFERENT TESTED METHODS WITH A DIFFERENT NUMBERS OF LABELED SAMPLES FOR THE AVIRIS INDIAN PINES SCENE. BOTH THE TOTAL NUMBER OF SAMPLES AND THE NUMBER OF SAMPLES PER CLASS ARE GIVEN (IN THE PARENTHESES)

Samples	RBF-SVM	3DCNN	SSRN	DBDA	A^2S^2K-ResNet	LinkNet	UNet	MSTNet
160 (10)	57.57%	56.38%	70.34%	72.82%	71.53%	57.46%	71.77%	**87.20%**
320 (20)	62.08%	63.53%	73.28%	79.52%	75.12%	75.39%	81.20%	**92.83%**
444 (30)	64.46%	70.67%	81.53%	84.81%	83.89%	84.95%	88.34%	**93.33%**
584 (40)	68.30%	74.12%	86.31%	87.24%	87.39%	89.62%	92.83%	**94.57%**
697 (50)	71.49%	79.24%	87.69%	88.07%	88.87%	92.71%	93.24%	**97.18%**

TABLE III

OVERALL, AVERAGE, K STATISTIC, AND INDIVIDUAL CLASS ACCURACIES FOR THE PAVIA UNIVERSITY DATA WITH 50 TRAINING SAMPLES PER CLASS. THE HIGHEST ACCURACIES ARE HIGHLIGHTED IN BOLD

Class	RBF-SVM	3DCNN	SSRN	DBDA	A2S2K-ResNet	LinkNet	UNet	MSTNet
1	88.05%	89.65%	97.16%	97.63%	96.33%	98.50%	**99.89%**	99.62%
2	90.34%	88.15%	95.11%	95.55%	94.95%	99.39%	**99.70%**	99.23%
3	71.75%	88.59%	93.89%	93.64%	92.13%	86.97%	95.79%	**99.43%**
4	92.11%	92.64%	97.53%	97.16%	**98.96%**	96.78%	98.35%	98.05%
5	98.25%	99.56%	**100.00%**	99.85%	**100.00%**	99.78%	99.78%	99.78%
6	73.05%	72.86%	97.57%	98.74%	96.13%	96.59%	99.49%	**99.98%**
7	78.22%	79.38%	99.47%	99.18%	99.08%	89.76%	**99.85%**	99.32%
8	80.11%	90.47%	96.41%	96.91%	**98.23%**	95.78%	93.16%	98.12%
9	99.90%	98.88%	**100.00%**	99.63%	**100.00%**	98.73%	98.96%	96.87%
OA	85.94%	85.99%	94.44%	94.81%	94.19%	97.43%	98.80%	**99.18%**
AA	85.75%	88.91%	97.46%	97.59%	97.31%	95.81%	98.33%	**98.93%**
Kappa	81.75%	81.93%	92.79%	93.26%	92.53%	96.61%	98.41%	**98.91%**

TABLE IV

OVERALL CLASSIFICATION ACCURACIES (IN PERCENTAGE) OBTAINED BY THE DIFFERENT TESTED METHODS WITH A DIFFERENT NUMBERS OF LABELED SAMPLES FOR THE ROSIS PAVIA UNIVERSITY SCENE. BOTH THE TOTAL NUMBER OF SAMPLES AND THE NUMBER OF SAMPLES PER CLASS ARE GIVEN (IN THE PARENTHESES)

Samples	RBF-SVM	3DCNN	SSRN	DBDA	A^2S^2K-ResNet	LinkNet	UNet	MSTNet
90 (10)	72.94%	79.00%	85.16%	88.13%	86.70%	67.27%	82.95%	**92.81%**
180 (20)	78.48%	83.84%	92.92%	89.37%	92.10%	92.65%	93.49%	**96.70%**
270 (30)	84.00%	84.36%	89.17%	92.54%	91.34%	93.60%	96.56%	**97.50%**
360 (40)	77.63%	83.14%	93.08%	93.34%	94.23%	96.27%	98.10%	**98.75%**
450 (50)	85.94%	85.99%	94.44%	94.81%	94.19%	97.43%	98.80%	**99.18%**

randomly as training data with a total of 697 samples (which constitutes approximately 6.80% of the available labeled samples). The overall accuracy (OA), average accuracy (AA), and kappa coefficient (k) were used to assess the performance of different classification methods. Table I displays the individual classification results of different methods, and their corresponding classification maps are shown in Fig. 7. Based on these results, the following preliminary conclusions can be drawn.

1) Compared with RBF-SVM, patch-based classifiers (e.g., 3DCNN, SSRN, DBDA, and A2S2K-ResNet) provided higher classification accuracy, which prove the efficiency of joint spectral information and spatial information. On the other hand, the above results also confirm that the model based on deep learning is more suitable for HSIC.
2) Compared with patch-based classifiers, image-based classifiers (e.g., LinkNet, UNet, and MSTNet) bring

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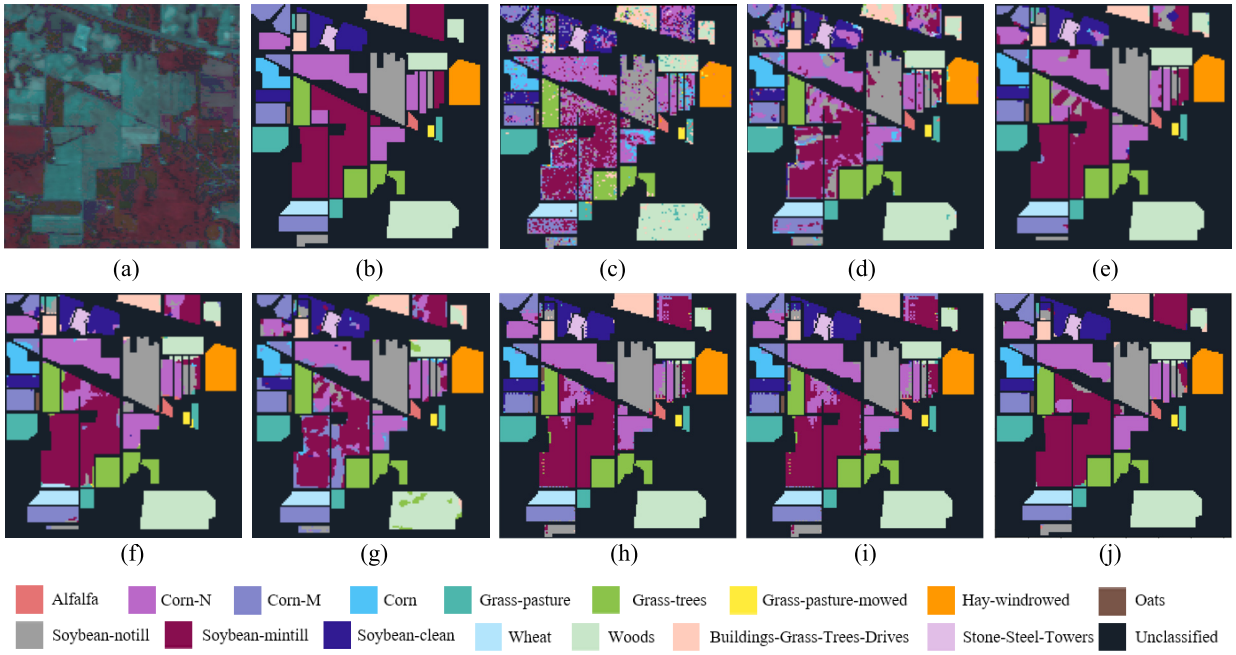


Fig. 7. Classification maps obtained by the different tested methods for the AVIRIS indian pines dataset. In all cases, 697 labeled samples in total (50 per class) were used. The overall classification accuracies are given in the parentheses. (a) True-color composite image. (b) Reference map. (c) RBF-SVM (71.49%). (d) 3DCNN (79.24%). (e) SSRN (87.69%). (f) DBDA (88.07%). (g) A^2S^2K-ResNet (88.87%). (h) LinkNet (92.71%). (i) UNet (93.24%). (j) MSTNet (97.18%).

TABLE V

OVERALL, AVERAGE, K STATISTIC, AND INDIVIDUAL CLASS ACCURACIES FOR THE SALINAS DATA WITH 50 TRAINING SAMPLES PER CLASS. THE HIGHEST ACCURACIES ARE HIGHLIGHTED IN BOLD

Class	RBF-SVM	3DCNN	SSRN	DBDA	A^2S^2K-ResNet	LinkNet	UNet	MSTNet
1	99.63%	98.15%	95.99%	95.88%	97.58%	**100.00%**	86.44%	97.52%
2	99.41%	99.35%	**100.00%**	99.95%	99.99%	98.96%	99.97%	**100.00%**
3	97.89%	95.31%	99.64%	98.91%	99.52%	97.62%	**100.00%**	**100.00%**
4	98.86%	**99.50%**	98.81%	98.73%	99.42%	98.56%	99.14%	99.06%
5	98.70%	96.61%	99.61%	99.32%	99.61%	98.55%	**99.66%**	99.48%
6	99.60%	98.69%	98.44%	98.42%	99.22%	99.62%	**100.00%**	99.37%
7	99.15%	98.13%	98.71%	98.70%	99.38%	97.88%	**100.00%**	99.97%
8	72.17%	82.58%	84.86%	86.75%	83.13%	99.55%	97.28%	**99.69%**
9	99.16%	98.29%	99.52%	99.51%	99.05%	98.04%	98.85%	**99.97%**
10	91.11%	91.57%	94.99%	93.87%	97.10%	98.21%	99.75%	**100.00%**
11	93.41%	86.44%	96.67%	96.48%	97.50%	95.60%	**100.00%**	96.72%
12	98.21%	96.58%	97.58%	97.50%	98.37%	96.32%	**100.00%**	98.19%
13	99.67%	95.43%	97.88%	97.83%	98.78%	95.11%	99.35%	**99.89%**
14	95.49%	97.40%	97.20%	97.01%	**98.02%**	94.76%	95.28%	96.57%
15	62.45%	70.24%	79.40%	78.90%	76.39%	97.03%	**99.99%**	99.63%
16	97.76%	93.56%	85.60%	85.63%	91.70%	97.27%	99.72%	**100.00%**
OA	87.87%	89.78%	90.82%	91.09%	91.28%	98.23%	98.53%	**99.49%**
AA	93.91%	93.61%	95.31%	95.21%	95.92%	97.69%	98.47%	**99.13%**
Kappa	86.53%	88.63%	89.85%	90.12%	90.34%	98.03%	98.36%	**99.43%**

an increase classification accuracy, which benefits the global information and demonstrates that image-based classification is a powerful framework for HSI.

3) Compared with UNet, MSTNet yielded better classification result, which indicates the improvement of exploration of global receptive field and the effectiveness

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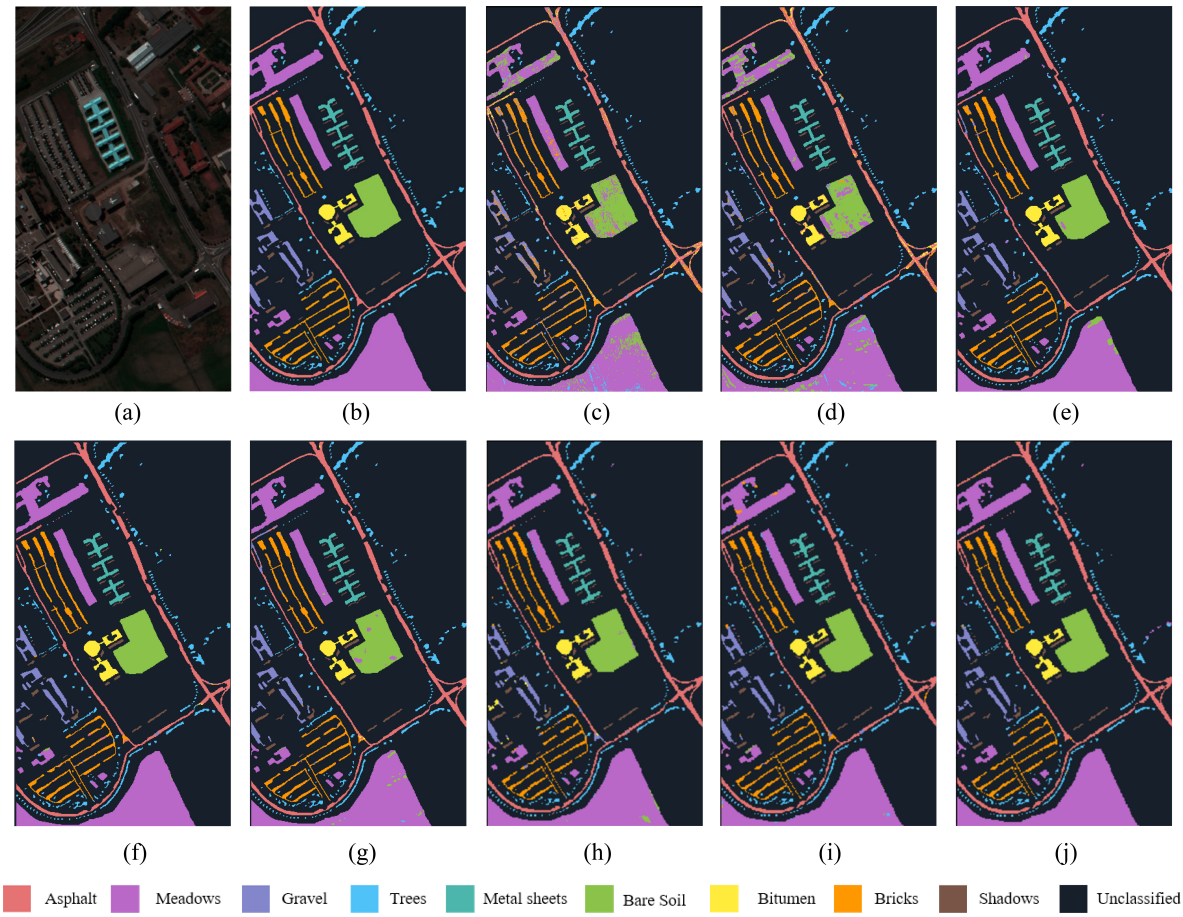


Fig. 8. Classification maps obtained by the different tested methods for the ROSIS university of pavia dataset. In all cases, 450 labeled samples in total (50 per class) were used. The overall classification accuracies are given in the parentheses. (a) True-color composite image. (b) Reference map. (c) RBF-SVM (85.94%). (d) 3DCNN (85.99%). (e) SSRN (92.05%). (f) DBDA (92.69%). (g) A^2S^2K-ResNet (94.19%). (h) LinkNet (97.43%%). (i) UNet (98.80%). (j) MSTNet (99.18%).

of feature extraction through transformer. On the one hand, the above results confirm that multilevel feature aggregation brings better representation.

In general, MSTNet obtains the best classification result in comparison with the other related tested methods, with an OA of 97.18%, which is 4.47% and 3.94% higher than LinkNet and UNet. Besides, MSTNet brings a better expression of details within the classification map, which proves the effectiveness of the proposed method.

In the second test with the Indian Pines scene, the tested methods are evaluated by a different number of labeled samples (from 10 to 50 samples per class). As shown in Table II, we can consolidate our previous conclusions. First, joint spectral information and spatial information are effective for HSIC, which can be seen from the results obtained by patch-based classifiers (e.g., 3DCNN, SSRN, DBDA, and A2S2K-ResNet). Second, the exploration of global information is conducive to obtain the detailed characteristics of HSI, which can be proved from image-based classifiers (e.g., LinkNet, UNet, and MSTNet). Finally, the designed TE and MFAD promote better characterization and representation, which can be verified from the improvement brought by MSTNet. In all cases, the proposed method obtained the best results in comparison with other models using image-based classification framework, which demonstrates its efficiency and stability.

2) University of Pavia Scene and Salinas Scene: In the first tests with the ROSIS University of Pavia Scene and the AVIRIS Salinas Scene, we randomly select 50 labeled samples per class with a total of 450 and 800 samples for training, while the remaining labeled samples are used for validation. Tables III and IV report the OA, AA, and kappa results of different tested methods, where Figs. 8 and 9 show the corresponding classification maps for two datasets, respectively. As shown in Tables III and IV, similar conclusions can be conducted with the Indian Pines scene's experiment. First, the patch-based classifiers (e.g., 3DCNN, SSRN, DBDA, and A2S2K-ResNet) achieve an increase in classification results relative to SVM by combining spectral information and spatial information. Then, the image-based classifiers (e.g., LinkNet, UNet, and MSTNet) also bring improvements in classification results compared with the patch-based classifiers by taking full advantage of global information. Finally, the proposed MSTNet obtains the best classification result with OA results of 99.18% and 99.49%, which is 0.38% and 0.96% higher than UNet for two datasets, respectively. In addition, it provides

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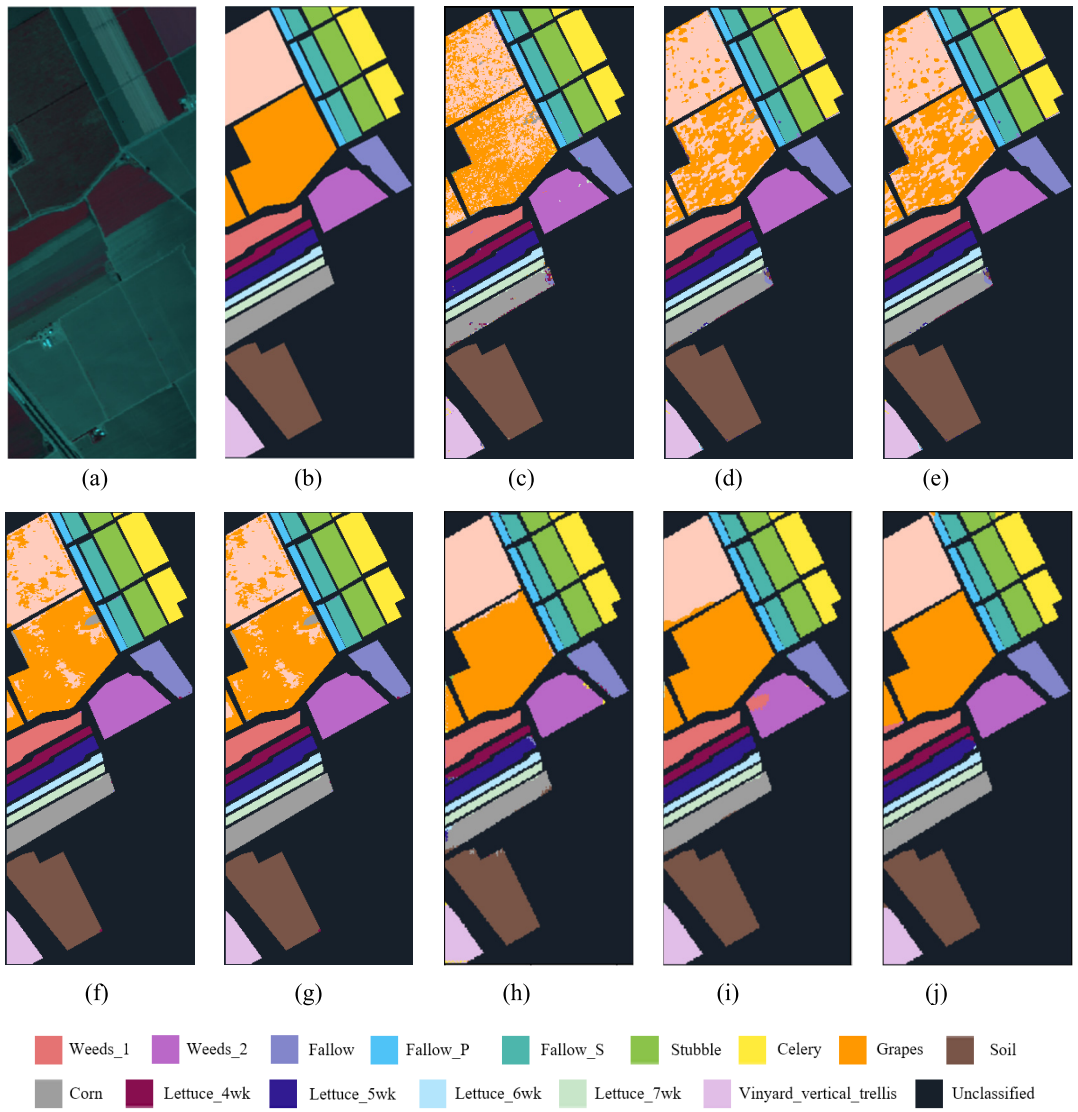


Fig. 9. Classification maps obtained by the different tested methods for the AVIRIS salinas dataset. In all cases, 800 labeled samples in total (50 per class) were used. The overall classification accuracies are given in the parentheses. (a) True-color composite image. (b) Reference map. (c) RBF-SVM (87.87%). (d) 3DCNN (88.58%). (e) SSRN (90.82%). (f) DBDA (91.09%). (g) A^2S^2K-ResNet (91.28%). (h) LinkNet (98.23%%). (i) UNet (98.53%). (j) MSTNet (99.49%).

better performances for individual classes, which can be seen from the classification maps shown in Figs. 8 and 9.

The second tests with the ROSIS University of Pavia scene and the AVIRIS Salinas scene evaluate the proposed MSTNet with a carrying size of labeled samples (from 10 to 50 samples per class). Tables V and VI show the OA results provided by different tested methods. As shown in Figs. 8 and 9, the superiority of the proposed method is confirmed by the best classification results obtained by MSTNet in all cases.

IV. Discussion

A. Evaluating Time Consumed for the Image-Based Classification Framework

Improving the time efficiency of CNN for HSI has always been a research hot spot. In order to compare the time consumption of two different frameworks, SSRN and MSTNet are chosen to represent patch-based frameworks and image-based frameworks, respectively. The batch size for SSRN is set to 32, and the batch size for MSTNet is set to 1 since the image-based classification processes an entire image at one time; 50 samples per class are randomly selected as training samples for each test. In addition, the patch size is set to two different sizes, including 7 and 11. Table VII shows the experimental results.

1) The training and testing time of MSTNet is not affected by the patch size since the entire image is fed into the network.
2) As the patch size increases, SSRN spends more training time than MSTNet in one epoch.
3) The results reveal SSRN lasted 95–240 times longer than MSTNet on test time.

It can be imagined that the patch-based classification method will consume more time since adjacent patches contain a lot

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TABLE VI

OVERALL CLASSIFICATION ACCURACIES (IN PERCENTAGE) OBTAINED BY THE DIFFERENT TESTED METHODS WITH A DIFFERENT NUMBERS OF LABELED SAMPLES FOR THE AVIRIS SALINAS SCENE. BOTH THE TOTAL NUMBER OF SAMPLES AND THE NUMBER OF SAMPLES PER CLASS ARE GIVEN (IN THE PARENTHESES)

Samples	RBF-SVM	3DCNN	SSRN	DBDA	A^2S^2K-ResNet	LinkNet	UNet	MSTNet
160 (10)	81.61%	81.13%	83.76%	80.89%	74.88%	73.18%	89.14%	**94.83%**
320 (20)	88.57%	84.37%	86.15%	89.41%	89.55%	94.28%	95.82%	**97.46%**
480 (30)	89.99%	86.65%	88.06%	83.01%	87.50%	95.13%	96.53%	**98.24%**
640 (40)	86.83%	88.70%	90.30%	90.72%	93.47%	97.14%	97.93%	**99.20%**
800 (50)	87.87%	89.78%	90.82%	91.09%	91.28%	98.23%	98.53%	**99.49%**

TABLE VII

TRAINING AND TESTING TIMES FOR PATCH-BASED AND IMAGE-BASED CLASSIFICATION

	Indian Pines			University of Pavia			Salinas		
	SSRN		MSTNet	SSRN		MSTNet	SSRN		MSTNet
Patch size	7	11	-	7	11	-	7	11	-
Train Time of One Epoch (s)	2.48	4.53	0.066	1.03	1.59	0.28	3.28	5.53	0.29
Test Time of One Epoch (s)	4.94	8.69	0.052	40.35	60.75	0.25	34.15	62.34	0.26

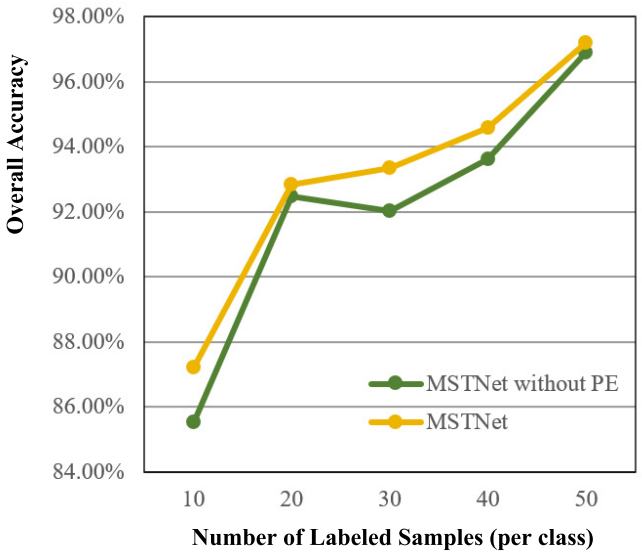


Fig. 10. Overall classification results obtained by the MSTNet without PE and the MSTNet as a function of the number of labeled (Per Class) for the AVIRIS indian pines dataset.

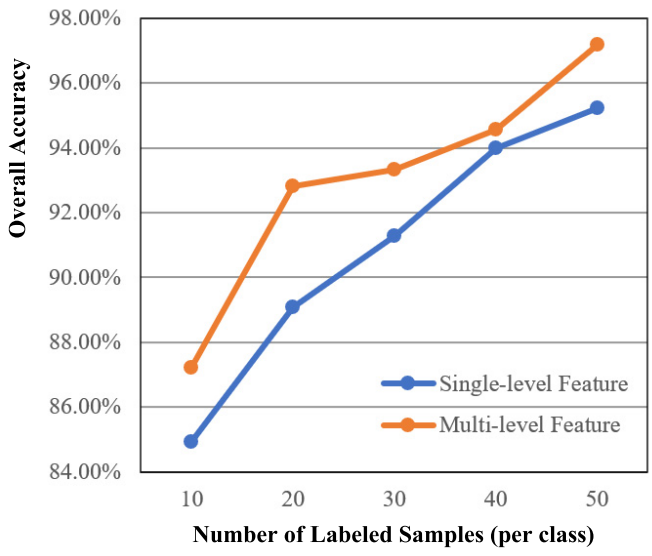


Fig. 11. Overall classification results obtained by the MSTNet with single-level feature and the MSTNet with multilevel feature as a function of the number of labeled (per class) for the AVIRIS indian pines dataset.

of redundant information when processing a larger image. Conversely, the image-based classification method is faster than the patch-based classification method because there is no redundant information in the test phase.

B. Evaluating the PE

It is significant for HSIC to utilize the spectral information and spatial information. In order to evaluate the effectiveness of the PE, ablation experiment with the AVIRIS Indian Pines Scene is set to compare MSTNet and MSTNet without PE with a carrying size of labeled samples (from 10 to 50 samples per class). Fig. 10 shows the experimental results. Compared with MSTNet without PE, MSTNet provides higher classification accuracy, which prove the efficiency of a learned PE in feature extraction.

C. Evaluating the MFAD

To validate the effectiveness of the MFAD, we compare the performance of MFAD and single-level feature decoder (SFD).

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In MFAD, multilevel features are connected together through channels to generate fusion features. Conversely, the last layer of features from TE is extracted in SFD. The experimental data selected AVIRIS Indian Pines Scene with a carrying size of labeled samples (from 10 to 50 samples per class).

As shown in Fig. 11, MFAD brings an increase in classification accuracy compared with the SFD, which indicates the improvement of integrated multilevel features. Generally, high-level features can produce more refined results, while low-level features are coarser. Combining the features is efficient by a top-down aggregation constructor through elementwise addition, which improve the interactions across different level features.

V. Conclusion

In this article, a novel MSTNet is proposed for HSIC. The main contributions of the proposed method first introduce an image-based classification framework to improve computing efficiency followed by the TE is used to extract features from global receptive field and can calculate the correlation between pixels, thereby improve classification accurate; finally, an MFAD is designed to integrate multilevel features, enhances interactivity among different features, and effectively improves the expressive ability of network. Experimental results on three real hyperspectral datasets demonstrate that the proposed MSTNet outperforms other related CNN-based methods for classification.

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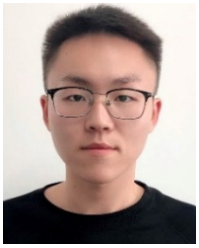
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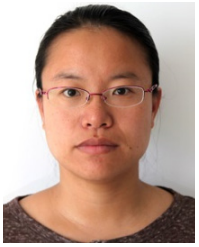
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